4.15

PAEDIATRICS

Paediatric Intensive Care Unit (PICU) Antimicrobial Guidelines

Version/Revision Date	Authors	Comments
1.0 (November 2021)	Paediatric Intensive Care Unit (PICU) Paediatric Surgical, and Paediatric Infectious Disease (ID) Teams	First release
2.0 (November 2024)	Paediatric Intensive Care Unit (PICU) Paediatric Surgical, and Paediatric Infectious Disease (ID) Teams	

Introduction

1. Nosocomial infections in PICU represent one of the major causes of morbidity and mortality in hospitalised patients. The overall mortality attributable to paediatric nosocomial infections has been estimated at 10-25% from the current literature review. Of these, ventilator-associated pneumonia (VAP), bloodstream infections (BSI), and urinary tract infections (UTIs) are the three commonly seen nosocomial infections in PICU. Skin infection, postoperative surgical wound infection, and gastrointestinal infections are some of the less frequently seen nosocomial infections.

2. Critically ill children are known to be vulnerable for PICU-acquired nosocomial infection due to suppression of both humoral and cellular immunity when they are ill, and the invasive procedures that clinicians put them through for treatment and monitoring. Therefore, in addition to the strict infection control surveillance and bundles care in PICU, a written guideline of antibiotics becomes essential, not just to have a standardised antibiotic practice, but also to help the clinicians to choose the most appropriate empiric antibiotics for the commonly

Introduction

encountered conditions in these group of children, aiming to optimise the clinical response to treatments, to minimise antibiotics over-usage and multidrug resistant infection problems, and ultimately to improve the morbidity and mortality of the critically ill patients.

Definitions

1. **Paediatric ICU-acquired infection:**
   An infection that was not present or incubating at the patient's admission to the ICU but became apparent during the ICU stay or within 48 hours after transfer from the ICU.

2. **Clinical sepsis (CS):**
   Bloodstream infection (BSI) with a negative culture but with at least one of the followings: fever or hypotension or oliguria, and a physician instituted treatment for sepsis, and there was no apparent infection at another site.

3. **Primary bloodstream infection or laboratory confirmed bloodstream infection (LCBSI):**
   One or more positive blood cultures, and the organism cultured from blood is not related to an infection at another site (no central line).

4. **Central line-associated bloodstream infection (CLABSI):**
   A positive blood culture with signs of systemic infection (fever, chills, and/or hypotension), with no apparent source of bacteraemia except the central venous catheter which has been in place for more than 48 hours.

5. **Ventilator-acquired pneumonia (VAP):**
   Nosocomial pneumonia that occurs in mechanically ventilated patients ≥ 48 hours after initiation of ventilatory support.

6. **Catheter-associated urinary tract infection (CAUTI):**
   $>10^5$ colonies of bacteria per ml with concomitant symptoms or pyuria.

7. **Surgical site infection (SSI)**:
   Superficial incisional, deep incisional, or organ/space: infections that occur within 30 days or 90 days after operative procedure with no implant, or within 1 year if an implant is placed and infection appears to be related to surgery.

   Please refer to the diagnostic criteria of VAP, CLABSI, CLAUTI & SSI at the appendix.

Principles of Antimicrobial Prescribing

1. Only prescribe antibiotics where there is clinical evidence of bacterial infection or a clear indication for prophylaxis.
2. Select the antibiotics to treat the likely underlying causative organisms based on
    - Clinical scenario: community, hospital or ICU acquired infection, any identifiable focus of infection, BSI or VAP, any in-dwelling lines (likes CVL, arterial line, urinary catheter, surgical drains etc.), any previous positive culture including colonisation with resistant organisms
    - Local antimicrobial susceptibility profile (antibiogram)
    - Severity of the infection
3. Ensure that appropriate microbiological specimens are taken prior to commencing antimicrobial treatment.
    - These may include blood, urine, nasopharyngeal aspirate (NPA) and BAL should be taken
    - Consider an LP, especially in infants, but this should not be done if there is any reduction in level of consciousness or any haemodynamic instability
    - Blood cultures should never be taken from a central line alone. Always take a clean peripheral sample (and a central line specimen if BSI sepsis is being considered)
4. The first dose of an intravenous antibiotic must be given within 60 minutes of prescription.
5. Empiric antibiotic choice should be reviewed once either Gram stain/microscopy/culture/sensitivity or PCR results (where appropriate) are available. Empiric therapy should be changed to directed therapy as soon as possible. Directed therapy should be the narrowest spectrum antibiotic to adequately cover the pathogens (e.g., If a patient has been empirically commenced on co-amoxiclav, and the culture isolate is sensitive to amoxicillin, co-amoxiclav should be changed to amoxicillin).
6. Intravenous antibiotics should be changed to an oral antibiotic as soon as clinically appropriate.
7. Avoid unnecessarily prolonged courses of antimicrobials.
8. In general, avoid topical use of antibiotics. Where topical use is indicated, do not use antibiotics that are used systemically.

Principles of Antimicrobial Prescribing

9. In general, avoid repeat courses with the same antibiotic.
10. Ensure any history of allergy is documented on the alert front sheet of the notes and on the prescription chart prior to commencing antibiotics.
11. Any dosing schedule should be in accordance with the standard Paediatric Drug Formulary (Frank Shann Drug Dose, BNF or Guy's and St. Thomas' Paediatric Formulary).
12. As a general rule, 5 - 7 days of treatment should suffice for most infections. In serious infections, the duration of therapy should be determined by the patient's condition and response to therapy.
13. Conditions requiring > 7 days of antibiotics, e.g., complicated pneumonia, brain abscess, acute exacerbation of cystic fibrosis, typhoid fever, infective endocarditis, intravascular endocarditis, meningitis, septic arthritis. Please refer to the hospital antibiotic guideline for further information.
14. It is expected that antibiotic prescriptions will be reviewed daily during ward rounds.
15. In the event that the patient does not respond within 24-48 hours of commencing treatment, ensure that dosage is correct, that infected foreign material is removed and pus is drained if indicated, and seek advice from the paediatric ID team.
16. Contact ID Team for advice on any aspect of antibiotic therapy.

CHAPTER 4.15	PAEDIATRIC INTENSIVE CARE UNIT (PICU) GUIDELINES		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Lower Respiratory Tract Infection/Pneumonia			
Community-acquired pneumonia (CAP)			

Infections and Likely Organisms	Suggested Treatment		Notes
	Recommended	Alternative	
Early onset: <48 hours after admission			
Streptococcus pneumoniae *Staphylococcus aureus* *Haemophilus influenzae*	Co-amoxiclav 25-30 mg/kg/dose (Amoxicillin componentI IV q8H	Ceftriaxone 50 mg/kg/dose IV q12H	
Ventilator-associated pneumonia (VAP) Late onset: >48 hours after admission			
Klebsiella pneumoniae *Pseudomonas aeruginosa*	Ceftazidime 50 mg/kg/dose IV q8H ± Gentamicin 4-5 mg/kg/dose IV q24H*	Cefepime 50 mg/kg/dose IV q8H **OR** Piperacillin/Tazobactam 100 mg/kg/dose (Piperacillin component) IV q8H	• *To monitor serum drug concentration of Gentamicin. • Caution with use of aminoglycoside in severely ill patients who may develop multiorgan dysfunction. • Prolonged use of broad-spectrum antibiotics i.e., Carbapenem, Piperacillin/Tazobactam, Cephalosporins is associated with emergence of *Stenotrophomonas maltophilia.*
Acinetobacter baumannii	Ampicillin/Sulbactam 50-100 mg/kg/dose (Ampicillin component) IV q6H	Ceftazidime 50 mg/kg/dose IV q8H	• Prolonged use of broad-spectrum antibiotics i.e., Carbapenem, Piperacillin/Tazobactam, Cephalosporins is associated with emergence of *Stenotrophomonas maltophilia.*
Stenotrophomonas maltophilia	Co-trimoxazole 5 mg/kg/dose (TMP component) q6-8H	Levofloxacin 8-10 mg/kg/dose PO q12H	• To check G6PD status prior to starting Co-trimoxazole.
Bloodstream Infection (BSI)			

CHAPTER 4.15	PAEDIATRIC INTENSIVE CARE UNIT (PICU) GUIDELINES		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Non-central line associated BSI (non-CLABSI)			
Escherichia coli *Streptococcus pneumoniae* Methicillin-sensitive *Staphylococcus aureus* (MSSA) Community-acquired Methicillin-resistant *Staphylococcus aureus* (CA-MRSA)	Co-amoxiclav 30 mg/kg/dose (Amoxicillin componentI IV q8H + Gentamicin 4-5 mg/kg/dose IV q24H*	Cloxacillin 50 mg/kg/dose IV q6H + Gentamicin 4-5 mg/kg/dose IV q24H*	• Always look for the source/primary site of infections and start antibiotics accordingly. • Community-acquired MRSA: Clindamycin 10 mg/kg/dose IV q8H • *To monitor serum drug concentration of Gentamicin.
Central line associated BSI (CLABSI)			
Methicillin-resistant Coagulase-negative *Staphylococci* (MRCoNS) *Acinetobacter baumannii* *Pseudomonas aeruginosa* *Klebsiella pneumoniae* Methicillin-resistant *Staphylococcus aureus* (MRSA)	Cefepime 50 mg/kg/dose IV q8H + Vancomycin 15 mg/kg/dose IV q8H* **In case of shock:** Meropenem 40 mg/kg/dose IV q8H +	Piperacillin/Tazobactam 100 mg/kg/dose (Piperacillin component) IV q8H + Vancomycin 15 mg/kg/dose IV q8H*	• *To monitor serum drug concentration of Vancomycin. • Remove CVL if possible. • Suspect vegetation or collection of pus if fever persists longer than 3 days despite treatment with the appropriate antibiotic based on the culture and sensitivity.

Bloodstream Infection

Antibiotic Adjustment

1. Once **antimicrobial susceptibility** results are available, de-escalation to specific and appropriate therapy is recommended. If blood cultures have no growth, the need for further empiric antibiotic treatment should be reassessed. If unexplained fever or sepsis persisted in a patient with a short-term central venous or arterial catheter and paired peripheral venepuncture and catheter blood cultures have failed to identify CLABSI, the catheter should be removed, and the tip should be sent for culture.

2. All **non-tunnelled catheters** causing CLABSI should be removed promptly, sometimes even before it is proven with the above criteria when clinical suspicion is high. Only for long-term catheters, salvage (systemic therapy coupled with antimicrobial lock [heparin + high concentration of antimicrobial agent that is selected based on susceptibility results]) can be attempted in limited instances such as:

    - Uncomplicated CLABSI caused by organisms other than Staphylococcus aureus, Pseudomonas aeruginosa, Bacillus spp, Micrococcus species, Propionibacteria, fungi, or mycobacteria.

    - The patients with limited vascular access sites or the ones who are solely dependent on central access for survival.

3. In the case of **tunnelled catheters**, uncomplicated exit site infections can be treated with short-course topical or systemic antimicrobials. All tunnel infections will require catheter removal due to a high risk of CLABSI. Additional indications for removal tunnelled catheters in CLABSI include:

    - Persistent symptoms > 36 hours or severe sepsis, hemodynamic instability(shock), and metastatic infection.

Bloodstream Infection

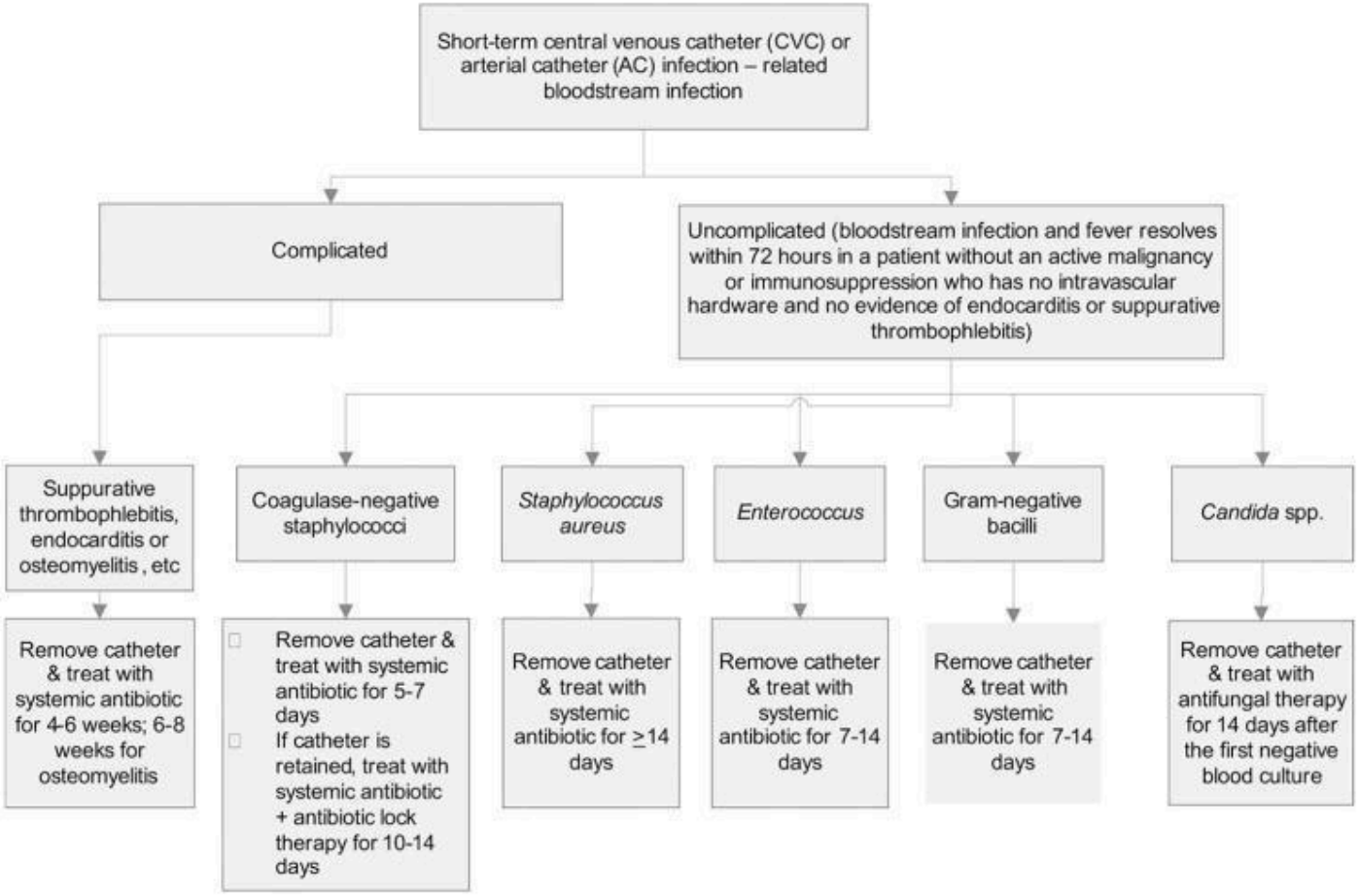
Antibiotic Adjustment

- Blood cultures that remain positive > 72 hours of appropriate antimicrobial therapy.
- Difficult to clear organisms (S. aureus, Pseudomonas, or fungi).
- Recurrence of uncomplicated CLABSI (any non-virulent organism) when salvaged for difficult access.

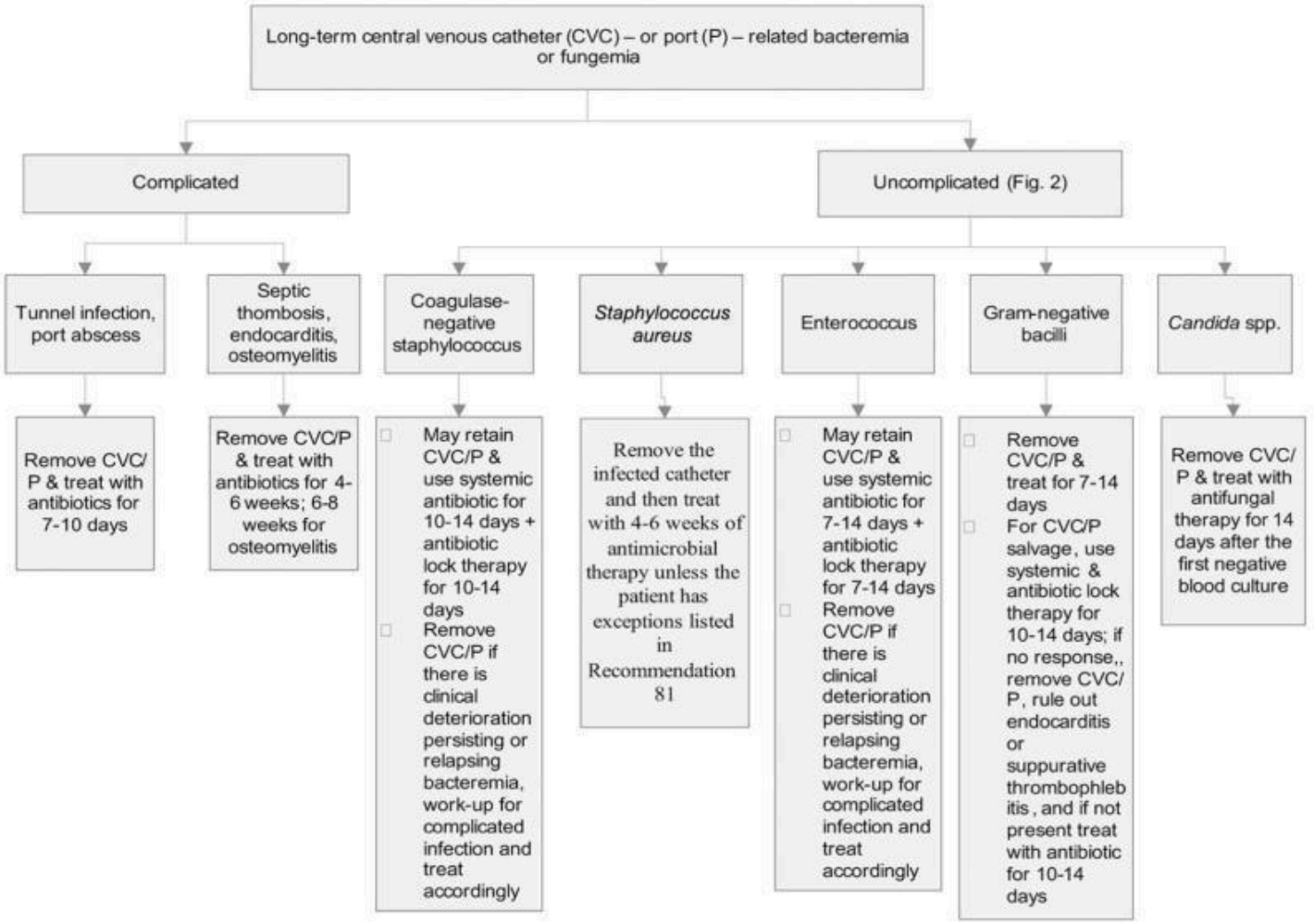
4. **Duration of therapy:** For patients with uncomplicated CLABSI (e.g., no endocarditis, or metastatic infection), in the absence of risk factors for hematogenous spread,(e.g., hardware, immunosuppression) and negative blood cultures 72 hours of catheter removal, following approach for intravenous antimicrobial therapy is recommended:

   - *Staphylococcus aureus*: 14 days in the absence of endocarditis
   - Coagulase-negative Staphylococci: 7 days
   - Enterococci and gram-negative bacilli: 10 to 14 days
   - Candida: 14 days in the absence of retinitis

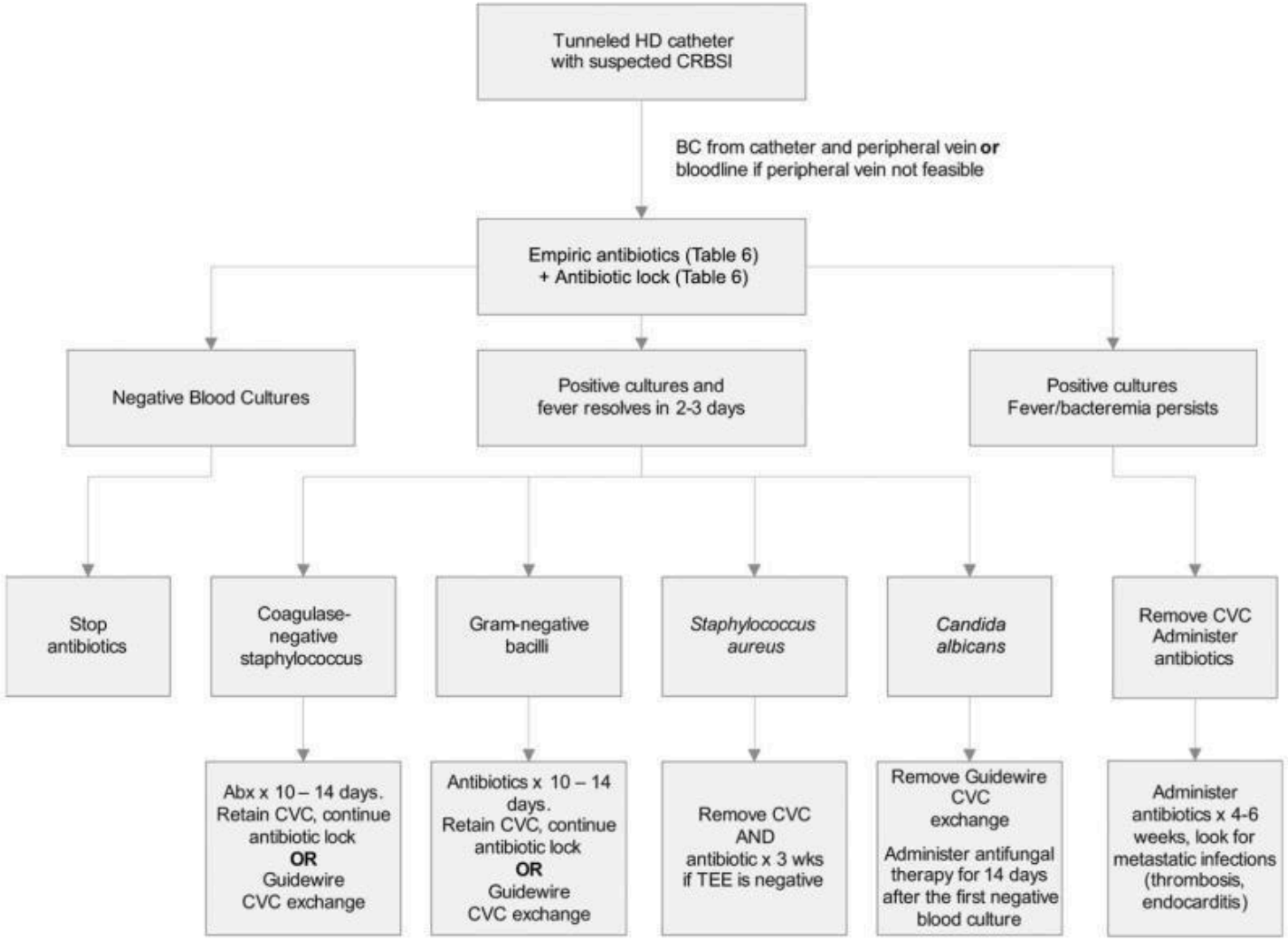
Approach to short-term CVL CLABSI



Approach to long-term CVL CLABSI



Approach to CLABSI Among Patients with a Tunnelled Haemodialysis Catheter



***Recommendation:** Patients can be considered for a shorter duration of antimicrobial therapy (i.e., a minimum of 14 days of therapy) if the patient is not diabetic; if the patient is not immunosuppressed (i.e., not receiving systemic steroids or other immunosuppressive drugs, such as those used for transplantation, and is non neutropenic); if the infected catheter is removed; if the patient has no prosthetic intravascular device (e.g., pacemaker or recently placed vascular graft); if there is no evidence of endocarditis or suppurative thrombophlebitis on TEE and ultrasound, respectively; if fever and bacteraemia*

resolve within 72 h after initiation of appropriate antimicrobial therapy; and if there is no evidence of metastatic infection on physical examination and sign- or symptom-directed diagnostic tests.

CHAPTER 4.15	PAEDIATRIC INTENSIVE CARE UNIT (PICU) GUIDELINES		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Catheter-associated Urinary Tract Infection (CAUTI)			
Escherichia coli *Pseudomonas aeruginosa* *Klebsiella pneumoniae* *Proteus mirabilis*	Co-amoxiclav 25-30 mg/kg/dose (Amoxicillin componentI IV q8H	Cefuroxime 25-50 mg/kg/dose IV q8H **OR** Cefalexin 25 mg/kg/dose **<7 days:** PO q12H **7-21 days:** PO q8H **21-28 days:** PO q6H **>28 days:** 12.5-25 mg/kg/dose PO q6-12H	• Remove urinary catheter if not indicated.
Yeast	Remove/change catheter immediately and re-culture urine under aseptic technique.		

Infections and Likely Organisms	Suggested Treatment		Notes
	Recommended	**Alternative**	
	If persistent growth, consider Fluconazole IV.		
Surgical Site Infection (SSI)			
Superficial (e.g., uncomplicated cellulitis)	Dressing Suture removal plus incision and drainage if appropriate Use antibiotics **IF** there is systemic symptoms OR raised WBC, erythema > 5 cm around incision, induration / necrosis	Cefalexin 25 mg/kg/dose PO q6-12H **OR** Cefuroxime 15 mg/kg/dose PO q12H **OR** Cloxacillin 15-25 mg/kg/dose PO q6H	• Final decision should be based on C&S. • If MRSA is suspected, Clindamycin 10 mg/kg/dose PO q8H **OR** Co-trimoxazole 5 mg/kg/dose (TMP dose) PO q12H.
Deep incisional (e.g., infected wound gaping)	Anatomical source control Cefuroxime 15 mg/kg/dose PO q12H **OR** 25 mg/kg/dose IV q8H	Cloxacillin 15-25 mg/kg PO q6H **OR** 50 mg/kg/dose IV q6H **OR** Cefuroxime 10-15 mg/kg/dose PO q12H **OR** 25 mg/kg/dose IV q8H **+**	• If community-acquired MRSA is suspected: Clindamycin 10 mg/kg/dose PO q8H **OR** Co-trimoxazole 5 mg/kg/dose (TMP dose) PO q12H. • If hospital-acquired MRSA is suspected: Vancomycin 15 mg/kg/dose IV q8H • Collect deep samples for C&S.

Infections and Likely Organisms	Suggested Treatment		Notes
	Recommended	Alternative	
	Wound of perineum, GIT, female genital tract: Co-amoxiclav 25 mg/kg/dose PO q12H **OR** IV q8H	Metronidazole 7.5 mg/kg/dose PO/IV q8H	• Antibiotics should be streamlined based on intraoperative culture results.
Organ/space (e.g., cuff cellulitis, intraabdominal abscess, pelvic abscess)	Anatomical source control. Follow C&S recommendations if available. **Empirical antibiotics:** Cefepime 50 mg/kg/dose IV q8H + Vancomycin 15 mg/kg/dose IV q8H	Piperacillin/Tazobactam 100 mg/kg/dose (Piperacillin component) IV q8H **OR** Meropenem 40 mg/kg/dose IV q8H + Vancomycin 15 mg/kg/dose IV q8H	• If community-acquired MRSA is suspected: Clindamycin 10 mg/kg/dose PO q8H **OR** Co-trimoxazole 5 mg/kg/dose (TMP dose) PO q12H. • If hospital-acquired MRSA is suspected: Vancomycin 15 mg/kg/dose IV q8H • Collect deep samples for C&S. • Antibiotics should be streamlined based on intraoperative culture results.

Appendix

Diagnosis of PICU-Acquired Infections

Ventilator-associated pneumonia (VAP)

For any patient

At least one of the following:

- Fever (>38°C)
- Leucopenia (<4000 WBC/mm3) or leukocytosis (>12,000 WBC/mm3)

And at least two of the following:

- New onset of purulent sputum (Tablan et al., 2004), or change in character of sputum, or increased respiratory secretions, or increased suctioning requirements
- New onset or worsening cough, or dyspnoea, or tachypnoea (Edwards et al., 2009)
- Rales or bronchial breath sounds
- Worsening gas exchange (e.g., O_2 desaturations (e.g., PaO_2/FiO_2 <300 or OI > 4), increased oxygen requirements, or increased ventilator demand)

Alternate criteria: For infants <1 years

Worsening gas exchange (e.g., O_2 desaturations [e.g., pulse oximetry <94%], increased oxygen requirements, or increased ventilator demand)

And at least three of the following:

- Temperature instability
- Leukopenia (<4000 WBC/mm3) or leukocytosis (>15,000 WBC/mm^3) and left shift (>10% band forms)

Ventilator-associated pneumonia (VAP)

- New onset of purulent sputum (Tablan et al., 2004) or change in character of sputum, or increased respiratory secretions or increased suctioning requirements
- Apnoea, tachypnoea (Edwards et al., 2009), nasal flaring with retraction of chest wall or grunting
- Wheezing, rales, or rhonchi
- Cough
- Bradycardia (<100 beats/min) or tachycardia (>170 beats/min)

Alternate criteria: For child >1 years or ≤12 years

At least three of the following:

- Fever (>38.4°C or >101.1°F) or hypothermia (<36.5°C or <97.7°F)
- Leukopenia (<4000 WBC/mm3) or leukocytosis (≥15,000 WBC/mm3)
- New onset of purulent sputum (Tablan et al., 2004), or change in character of sputum (Edwards et al., 2009), or increased respiratory secretions, or increased suctioning requirements
- New onset or worsening cough, or dyspnoea, apnoea, or tachypnoea (Edwards et al., 2009).
- Rales or bronchial breath sounds
- Worsening gas exchange (e.g., O_2 desaturations [e.g., pulse oximetry <94%], increased oxygen requirements, or increased ventilator demand)

Radiography

Two or more serial chest radiographs with at least one of the following (Klevens et al., 2007) (Dudeck et al., 2011):

- New or progressive and persistent infiltrate
- Consolidation

Ventilator-associated pneumonia (VAP)

- Cavitation
- Pneumatoceles, in infants ≤1-year-old

Note: *In patients without underlying pulmonary or cardiac disease (e.g., respiratory distress syndrome, bronchopulmonary dysplasia, pulmonary oedema, or chronic obstructive pulmonary disease), one definitive chest radiograph is acceptable).*

Laboratory

At least ONE of the following:

- Positive growth in blood culture not related to another source of infection
- Positive growth in culture of pleural fluid
- Positive quantitative culture from minimally-contaminated LRT specimen (e.g., BAL or protected specimen brushing)
- Bronchoscopy BAL: ≥ 10^4 cfu/ml
- Modified BAL: ≥ 10^5 cfu/ml

Central line-associated bloodstream infection (CLABSI)

Must meet one of the following criteria

Criteria 1

Patient has a recognized pathogen cultured from one or more blood cultures

AND

Organism cultured from blood is not related to an infection at another site

Central line-associated bloodstream infection (CLABSI)

Criteria 2

Patient has at least one of the following signs or symptoms: fever (>38°C), chills, or hypotension

AND

Positive laboratory results are not related to an infection at another site

AND

The same common commensal (i.e., diphtheroids [*Corynebacterium spp.* not *C. diphtheriae*], *Bacillus spp.* [not *B. anthracis*], *Propionibacterium spp., coagulase-negative staphylococci [including S. epidermidis], viridans group streptococci, Aerococcus spp., and* Micrococcus spp.) is cultured from two or more blood cultures drawn on separate occasions (< 24 hours apart)

Criteria 3

Patient ≤ 1 year of age has at least one of the following signs or symptoms: fever (>38°C core), hypothermia (<36°C core), apnea, or bradycardia

AND

Positive laboratory results are not related to an infection at another site

AND

The same common commensal (i.e., diphtheroids [*Corynebacterium spp.* not *C. diphtheriae*], *Bacillus spp.* [not *B. anthracis*], *Propionibacterium spp., coagulase-negative staphylococci [including S. epidermidis], viridans group streptococci, Aerococcus spp., and* Micrococcus spp.) is cultured from two or more blood cultures drawn on separate occasions (< 24 hours apart)

Catheter-associated urinary tract infection (CAUTI)

Patient must meet a, b, and c below:

a. Indwelling urinary catheter (IUC) that had been in place for more than 2 consecutive days (> 2 calendar days) in an inpatient location on the date of event AND was either:

- present for any portion of the calendar day on the date of event, **OR**

Catheter-associated urinary tract infection (CAUTI)

- removed the day before the date of event

b. At least one of the following signs or symptoms:

- fever (>38.0°C)
- suprapubic tenderness
- costovertebral angle pain or tenderness
- urinary urgency
- urinary frequency
- dysuria

c. The patient has a urine culture with no more than two species of organisms identified, at least one of which is a bacterium of ≥105 CFU/ml. Mixed flora, yeast, mould, fungus (Candida species) or parasites cannot be used to meet the UTI definition.

Surgical site infection (SSI)

A surgical site infection (SSI) is an infection that occurs AFTER surgery in the part of the body where the surgery took place. This is either a new infection or an infection caused by treatment failure within 30 days or 90 days after operative procedure with no implant, or within 1 year if an implant is placed and infection appears to be related to surgery.

SSI is a difficult term to define accurately because it has a wide spectrum of possible clinical features. US Centers for Disease Control and Prevention (CDC) classified SSI into incisional SSIs, which can be superficial or deep, and organ/space SSIs, which affect the rest of the body other than the body wall layer. Superficial incisional SSI is more common than deep incisional SSI and organ/space SSI.

Definitions

Surgical site infection (SSI)

When surgical site infection is suspected by the presence of cellulitis, either by a new infection or an infection caused by treatment failure, SSI should be suspected. Three types of SSI are **superficial**, **deep incisional**, and **organ/space**.

Superficial

Involves skin or subcutaneous tissue of the incision and rarely lead to systemic inflammatory response

Patient has **at least one** of the following:

- purulent drainage from the superficial incision.
- organism(s) identified from an aseptically-obtained specimen from the superficial incision or subcutaneous tissue by a culture or non-culture based microbiological testing method
- superficial incision that is deliberately opened by a surgeon, physician* or physician designee and culture or non-culture based testing of the superficial incision or subcutaneous tissue is not performed

  AND

  patient has at least one of the following signs or symptoms: localised pain or tenderness; localised swelling; erythema; or heat
- diagnosis of a superficial incisional SSI by a physician/surgeon

Deep incisional

Includes tissues down to including fascia and muscle

Patient has **at least one** of the following:

- purulent drainage from the deep incision
- a deep incision that spontaneously dehisces, or is deliberately opened or aspirated by physician/surgeon

  AND

Surgical site infection (SSI)

organism(s) identified from the deep soft tissues of the incision by a culture or non-culture-based microbiological testing method. A culture or non-culture based test from the deep soft tissues of the incision that has a negative finding does not meet this criterion

AND

patient has at least one of the following signs or symptoms: fever

- (>38°C); localized pain or tenderness
- an abscess or other evidence of infection involving the deep incision that is detected on gross anatomical or histopathologic exam, or imaging test

Organ/space

Include any body cavity that was opened or manipulated during surgery

Patient has **at least one** of the following:

- purulent drainage from a drain that is placed into the organ/space (for example, closed suction drainage system, open drain, T-tube drain, CT-guided drainage)
- organism(s) identified from fluid or tissue in the organ/space by a culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (for example, not Active Surveillance Culture/Testing (ASC/AST))
- an abscess or other evidence of infection involving the organ/space that is detected on gross anatomical or histopathologic exam, or imaging test evidence suggestive of infection.

References:

1. Pneumonia (Ventilator-associated [VAP] and non-ventilator-associated Pneumonia [PNEU]) Event. Device-associated Module PNEU Events, January 2021. www.cdc.gov › pdfs › pscmanual › 6pscvapcurrent
2. Bloodstream Infection Event (Central Line-Associated Bloodstream Infection and Non-central Line Associated Bloodstream Infection). Device-associated Module BSI, January 2021. www.cdc.gov › nhsn › pdfs › pscmanual.

3. Urinary Tract Infection (Catheter-Associated Urinary Tract Infection [CAUTI] and Non-Catheter-Associated Urinary Tract Infection [UTI]) Events. Device-associated Module UTI, January 2021. www.cdc.gov › pdfs › pscmanual › 7psccauticurrent
4. Surgical Site Infection Event (SSI). Device-associated Module SSI, January 2021. www.cdc.gov › nhsn › pdfs › pscmanual› 9pscssicurrent
5. Clinical Practice Guidelines for the Diagnosis and Management of Intravascular Catheter-Related Infection: 2009 Update by the Infectious Diseases Society of America, IDSA guideline